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Multiscale parallel feature aggregation network with attention fusion (MPFAN-AF): A novel approach to cataract disease classification

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Author contributions
All authors contributed to the study's conception and design. Material preparation, data collection, and analysis were carried out by Ansari MA, Mewada A, and Ahmad S. The first draft of the manuscript was written by Ansari MA, and all authors provided feedback on previous versions of the manuscript. All authors read and approved the final version of the manuscript.

Competing interests
The authors declare no conflicts of interest.

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Abbreviations
DL, deep learning; CNNs, convolutional neural networks; MPFAN-AF, multiscale parallel feature aggregation network with attention fusion; Conv2D, 2D convolutional layers; FPR, false positive rate.

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Abstract

Background: Early and accurate diagnosis of cataracts, which ranks among the leading preventable causes of blindness, is critical to securing positive outcomes for patients. Recently, eye image analyses have used deep learning (DL) approaches to automate cataract classification more precisely, leading to the development of the Multiscale Parallel Feature Aggregation Network with Attention Fusion (MPFAN-AF). Focused on improving a model's performance, this approach applies multiscale feature extraction, parallel feature fusion, along with attention-based fusion to sharpen its focus on salient features, which are crucial in detecting cataracts. **Methods:** Coarse-level features are captured through the application of convolutional layers, and these features undergo refinement through layered kernels of varying sizes. Moreover, this method captures all the diverse representations of cataracts accurately by parallel feature aggregation. Utilizing the Cataract Eye Dataset available on Kaggle, containing 612 labelled images of eyes with and without cataracts proportionately (normal vs. pathological), this model was trained and tested. **Results:** Results using the proposed model reflect greater precision over traditional convolutional neural networks (CNNs) models, achieving a classification accuracy of 97.52%. Additionally, the model demonstrated exceptional performance in classification tasks. The ablation studies validated that all applications added value to the prediction process, particularly emphasizing the attention fusion module. **Conclusion:** The MPFAN-AF model demonstrates high efficiency together with interpretability because it shows promise as an integration solution for real-time mobile cataract detection screening systems. Standard performance indicators indicate that AI-based ophthalmology tools have a promising future for use in remote conditions that lack medical resources.

Keywords: cataract classification; deep learning; multiscale feature extraction; attention mechanism; medical image analysis

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Introduction

Cataracts are one of the leading causes of visual impairment and blindness globally, especially affecting the elderly population [1, 2]. Cataracts account for approximately 51% of global blindness, affecting over 65 million people, according to the World Health Organization [3]. The burden is disproportionately high in low- and middle-income countries, where access to timely diagnosis and surgical intervention is limited. This leads to preventable vision loss, reduced quality of life, and economic hardship due to loss of productivity and increased dependence. The condition is characterized by clouding of the lens in the eye, leading to progressive vision deterioration, as shown in Figure 1. Timely diagnosis and intervention are essential to preventing permanent vision loss.

Traditionally, diagnosing cataracts relies heavily on clinical evaluations and medical professionals' manual inspection of ophthalmic images, which may be subject to variability and require significant time and expertise. With the growing demand for faster and more accessible diagnostic systems, artificial intelligence and deep learning (DL) models have emerged as promising alternatives for automating cataract detection with high accuracy and reliability [2, 4]. The last few years have experienced a progressive surge in the usage of convolutional neural networks (CNNs) for classifying medical images, including detecting eye diseases using DL. Although cataracts present distinct eye features that challenge more traditional CNN models, one of the traditional model's limitations includes fixed-sized receptive fields for convolution layers, preventing feature extraction at different levels of parts. Since features differ in their manifestation with respect to strength and spatial location with respect to low-level and high-level details, multi-dimensional structures are needed to distinguish them from other eye features.

In addition, most existing models lack sufficient focus on important areas within an image, which are crucial for the model, depending on context. The lack of attention hinders the model's reliability, especially when images are corrupted with noise, lighting, or occlusions due to real-world scenarios. Moreover, traditional architectures usually do not achieve a balance between capturing fine-grained local details and contextual information [5, 6]. The lack of ability to capture complex details from different patterns contributes to unsatisfactory classification output, especially for datasets with low inter-class differences, such as images of eyes with cataracts compared to normal eyes.

To address these problems, we developed a deep neural architecture, the Multiscale Parallel Feature Aggregation Network with Attention Fusion (MPFAN-AF). This model captures multiscale features with different convolutional filters of various kernel sizes. It also captures such features in parallel and uses attention fusion to improve them [7–9]. The attention mechanism enables the model to focus on specific regions of the image, which improves understanding of intrinsic patterns and accuracy. With this model design, MPFAN-AF combines different eye cataract pupils' images with normal pupils for efficient and precise classification.

The key contributions of this work are as follows:

1. Multiscale feature extraction using convolution kernels of various sizes to capture fine, medium, and coarse patterns in eye images.
2. Parallel feature aggregation to combine multiple scale-based features without loss of spatial information.
3. Attention fusion mechanism to selectively emphasize regions that are most indicative of cataract presence.
4. Extensive ablation and performance analysis demonstrating the superiority of the proposed model over existing CNN-based classifiers.

The remainder of this paper is organized as follows: the next part provides a comprehensive overview of related work, highlighting the strengths and limitations of existing methods for glaucoma detection using DL techniques. The following part presents the proposed MPFAN-AF architecture in detail, elaborating on each component, including multiscale feature extraction, parallel feature aggregation, and attention fusion mechanism. The subsequent part discusses the experimental setup, dataset characteristics, and evaluation metrics, followed by detailed performance analysis and comparison with state-of-the-art models. The final part concludes the paper with the main findings, implications for clinical deployment, and potential directions for future work.

Related work

In cataract classification, DL models, particularly CNNs, can evolve significantly from basic sequential architectures to more advanced networks. Early CNNs like LeNet-5, AlexNet, and VGGNet are pioneering efforts that laid the foundation for using DL in medical image classification [10–12]. LeNet-5, one of the earliest CNNs, moderately performed cataract classification but was limited by its simple architecture, which constrained its ability to manage complex features. With increased parameters and deeper layers, AlexNet and VGGNet showed significant improvements but struggled with fine details and different cataract severities.

Innovations in feature extraction and increasing network depth were seen with newer models such as DenseNet, ResNet, and InceptionNet [13–15]. DenseNet boosted model accuracy and efficiency by enhancing feature reuse thanks to its variant of densely connected networks. ResNet improved the performance of complex datasets with the use of residual connections, allowing for deeper networks without facing the vanishing gradient problem. InceptionNew's multi-branch structure was better suited towards cataract detection since it captured features of varying sizes and scales, which is advantageous where size and shape may differ. All these models achieved improved robustness and accuracy for classification, especially considering the wide range of severity and appearance differences among cataract images.

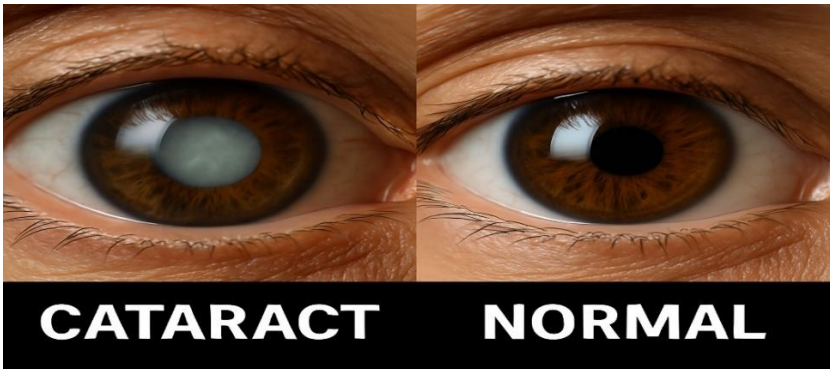


Figure 1 Visual comparison between cataract-affected and normal eye images.
Example images from the Cataract Image Dataset (Nandanp6. Cataract Image Dataset. Kaggle. Accessed March 15, 2025. Available at: https://www.kaggle.com/datasets/nandanp6/cataract-image-dataset).

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Hybrid architecture, containing CNN and Recurrent Neural Network along with multiscale processing and attention mechanisms, has been added to develop CNNs further [16, 17]. Attention-based CNNs improve model accuracy through relevance filtering within an image: identifying features that are instrumental to its identification [18]. From a different angle, multiscale CNNs are more efficient in clinical practice because they can identify cataracts at any stage, from their early (mild) to advanced (severe) stages, due to their varying configurations tailored towards different levels of complexity. The ability of the hybrid CNN-RNN model (Temporal-Spatial Encapsulation Model) to analyze multiple data streams simultaneously provides insight into revealing the eye disease while tracking its progression, making it easier to monitor disease progression over time [19]. Such complex systems today provide the utmost precision and adaptability in classifying cataracts. Kumar et al. utilized a hybrid neural network design to improve the precision of retinal disease classification within neural frameworks [20]. Their model incorporated convolutional, dense, and recurrent layers, which helped capture spatial and sequential features within retinal images. It performed well under several conditions, thus illustrating its value for reliable automated diagnostic systems in ophthalmology. Suryawanshi and Reddy proposed an advanced framework for detecting and classifying cataracts based on the contemporary YOLOv9 object detection algorithm combined with other machine learning classifiers [21]. Their work aims to resolve the still localization of cataract-affected areas while providing inference in real-time. The results exemplified remarkable accuracy along with rapid computation speeds, which are essential for point-of-care evaluation systems and screenings. This study demonstrates increasing use of object detection algorithms in analysis of ophthalmic images.

DL models have technical issues, as they need large, annotated datasets, and they have issues with generalizing to diverse clinical images. They also require a lot of computation capacity, which makes real-time implementation very challenging [22]. It is this kind of achievement that MPFAN-AF, in this case, tries to extend and address the limitation posed by these methods with a novel multi-parallel model of feature aggregation at different levels, increasing the resolution of the model by coarse and fine feature detection with a common construct. The attention fusion mechanism also enhances classification accuracy by enabling the model to ignore more advanced cataract signs and focus on the primary areas of interest with the most vital cataract details, which are critical for proper cataract-type classification. These two, in conjunction, enable MPFAN-AF to outperform traditional systems in efficiency and accuracy regarding varying image quality, levels of cataract severity, and traditional methods within the classification. Our system proves more suitable for clinical diagnosis because it demonstrates precision and adaptation capabilities toward various patient data samples.

Proposed methodology

MPFAN-AF aims to detect eye contract disease by using multiscale extraction of features, parallel aggregation of features, and attention fusion as the key methods. Figure 2 contains the diagram of the MPFAN-AF architecture. In this section, we explain the step-by-step approach for constructing the constituency structure in chronological order and mathematical formulas to improve the accuracy and robustness of classification.

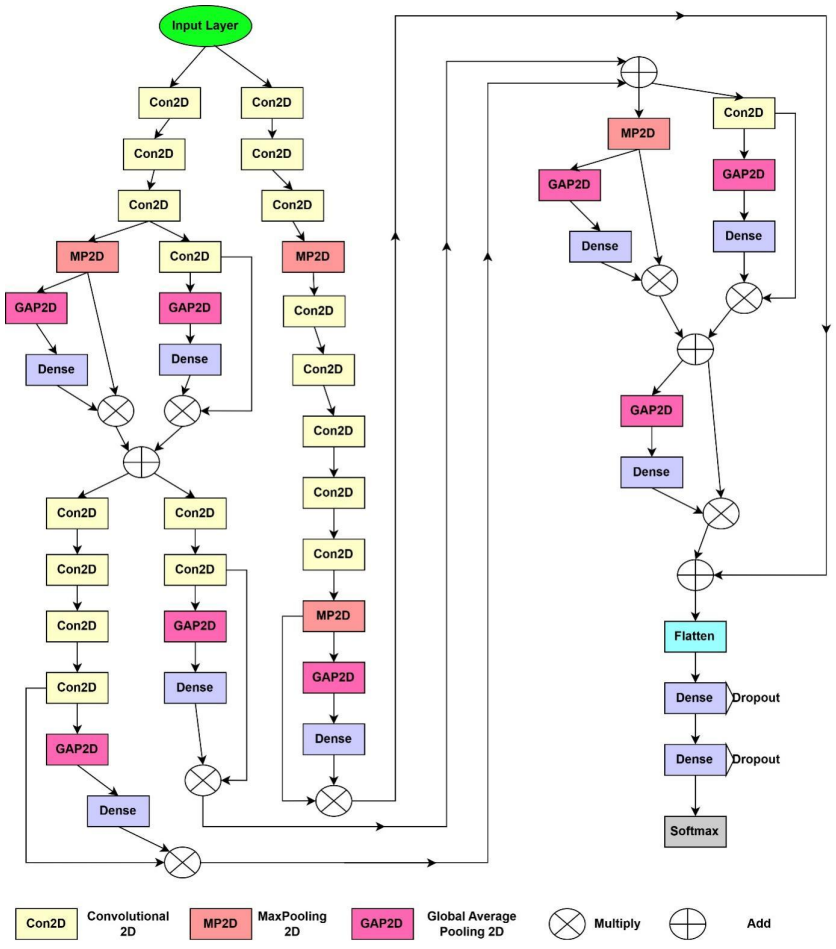


Figure 2 MPFAN-AF model architecture.
MPFAN-AF, multiscale parallel feature aggregation network with attention fusion.

The methodology incorporates a deep CNN with an attention-based fusion block on holistic feature extraction to enable effective combination and refinement of features at various abstraction levels. The network contains an input layer, which is made for 120×120 Red-Green-Blue channel enabled images, then 2D convolutional layers (Conv2D) with Rectified Linear Unit activations follow. These layers are designed to capture rudimentary features like edges, textures, and patterns. The architecture has several convolutional blocks, and each block consists of successive layers of convolutions of different kernel sizes, like 3×3 and 5×1, which help the network learn to represent features at multiscale vertically stacked within the same block. To downsample the feature maps, the MaxPooling2D layers reduce dimensions while spatially retaining the most salient information. Convolutional blocks are mixed with attention fusion mechanisms that enhance the most salient features from varying pathways using GlobalAveragePooling2D to compute context for each feature map and Dense layers to generate attention weights. The attention weights scale the feature maps via element-wise multiplication (Multiply layer), thus the network narrows its positions of focus to meaningful spatial areas and flattens irrelevant parts, as shown in Figure 3. This attention fusion is conducted in a hierarchical manner whereby the features are refined to narrow down features as they are finally combined, thereby enabling the model to learn progressively more discriminative representations at every step.

In the proposed attention fusion mechanism, attention weights are learned through global context extraction and a stepwise nonlinear process. First, spatially aggregated information is captured via global average pooling over the input feature maps. Following this, the pooled output is processed by a dense layer with sigmoid activation, resulting in scale-limited attention weights [0, 1] for each channel of the feature. Guided by the computed attention weights, the original feature maps are modified using elementwise multiplication, hence weighted interception. This operation dynamically scales feature maps to enable the model to enhance informative regions while suppressing irrelevant or noisy data streams within those regions. The model strengthens its internal representation of training inputs during training through attending to relevant spatial regions most predictive of cataracts. Such selective enhancement aids discriminative ability on features, which significantly improves the classification accuracy of MPFAN-AF.

The focus paradigm, when combined with particular attention mechanisms, becomes crucial at higher levels as it allows amalgamation of diverse features from several convolutional pathways from different layers together into a singular construct for further learning. For example, the network incorporates multiple pathways where different convolutional operations (e.g., 1×1 convolutions, 5×1 convolutions, and 3×3 convolutions) are applied to the feature maps, and the results are fused through attention mechanisms. The feature maps are then passed through additional pooling layers, such as 3×3 MaxPooling2D, which help to compress further and abstract the information, preserving the most informative features while reducing spatial complexity. After passing through the convolutional and pooling layers, the output is a feature map that is flattened into a one-dimensional vector containing high-level features that dense layers can process. The last dense layers use Rectified Linear Unit activations and are set to 1,024 and 256 units, then dropout layers are applied to counter overfitting. A softmax activation function using two output units for the class probabilities is then used for the final classification. The attention mechanism allows the model to identify important feature map segments, enhancing its performance on difficult image classification tasks through noise reduction and enhanced representation quality. The design features in the network, including deep layers, attention fusion, and multiscale features, make the model learn various levels of spatial relationships for superior accuracy and dataset generalization.

The design of the proposed MPFAN and attention feature fusion module has been captured in detail in Table 1 and 2. Table 1 presents a feature fusion method through an attention mechanism. Here, the attention mechanism multiplies the matrix output of global average pooling in two feature maps with a sigmoid dense layer channel-wise attention. It performs summation to yield a fused output that improves attention in features. Table 2 outlines the model architecture, beginning from the convolutional layers with initial feature extraction, multi-path parallel processing, attention fusion, and ending with sequential fusion. The model uses high-level deep hierarchical feature encoding, a second feature path stemming from the input that provides plausible complementary information. These final fused features undergo flattening, followed by passing through dense layers with a certain amount of dropout regularization and then yield classification output via softmax function layers. Attention-based multi-branch and other soft multi-branch networks are employed as focus fusion architectures to incorporate the attention-mechanism feature into model construction, to strengthen the feature learning, structure showcase framework through selective focus.

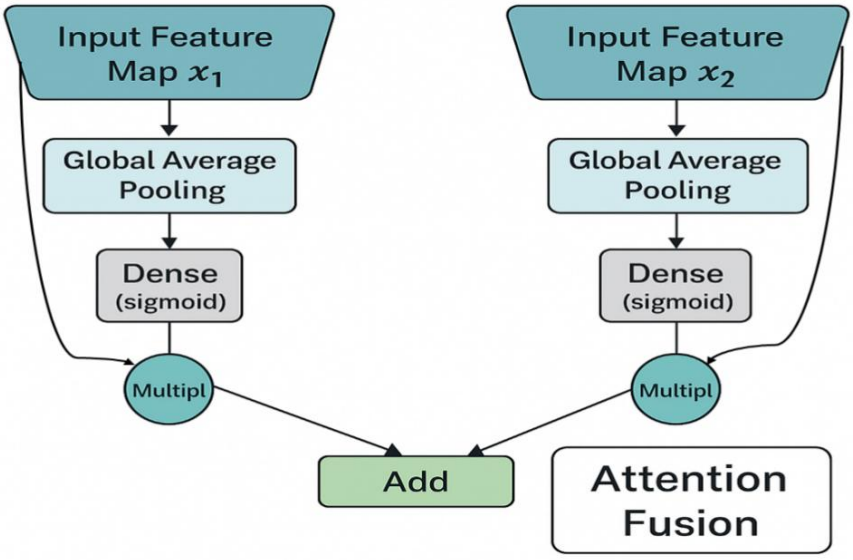


Figure 3 The architecture of the attention fusion module used in MPFAN-AF.
MPFAN-AF, multiscale parallel feature aggregation network with attention fusion.

Table 1 Algorithm for attention-based feature fusion

Step	Details
Function	AttentionFusion (x1, x2)
Input	x1, x2: Feature maps
Process	1. Compute global average pooling for $x1 \rightarrow g1 \leftarrow Global_Average_Pooling(x1)$ 2. Apply sigmoid dense layer to $g1 \rightarrow a1 \leftarrow Dense(g1, activation = 'sigmoid')$ 3. Multiply $x1$ with $a1 \rightarrow x1_weighted \leftarrow x1 * a1$ 4. Compute global average pooling for $x2 \rightarrow g2 \leftarrow Global_Average_Pooling(x2)$ 5. Apply sigmoid dense layer to $g2 \rightarrow a2 \leftarrow$ Dense(g2, activation = 'sigmoid') 6. Multiply $x2$ with a2 $\rightarrow x2_weighted \leftarrow x2 * a2$ 7. Fuse both $weighted\ features \rightarrow fused \leftarrow x1_weighted + x2_weighted$
Output	Attention-weighted fused feature map

Table 2 Algorithm for building proposed deep MPFAN-AF network

Step	Details
Function	Build Attention-CNN Model
Input	RGB image of shape (120, 120, 3)
Output	Classification output of shape (2,) with softmax activation
Step 1: Initial feature extraction	1. x ← Input(image) 2. x ← Conv2D(x, 32, 3x3, stride = 2, relu) 3. x ← Conv2D(x, 32, 3x3, stride = 1, relu) 4. x ← Conv2D(x, 64, 3x3, stride = 1, relu)
Step 2: First fusion	5. p1 ← MaxPool(x, 3x3, stride = 2) 6. x2 ← Conv2D(x, 64, 3x3, stride = 2, relu) 7. feat_1 ← AttentionFusion(p1, x2)
Step 3: Parallel paths from feat_1	Path A: 8. a1 ← Conv2D(feat_1, 64, 3x3, relu) 9. a1 ← Conv2D(a1, 96, 5x5, relu) Path B: 10. b1 ← Conv2D(feat_1, 64, 1x1, relu) 11. b1 ← Conv2D(b1, 64, 5x1, relu) 12. b1 ← Conv2D(b1, 64, 1x5, relu) 13. b1 ← Conv2D(b1, 96, 3x3, relu) 14. feat_2 ← AttentionFusion(a1, b1)
Step 4: Second fusion	15. p2 ← MaxPool(feat_2, 3x3, stride = 2) 16. x3 ← Conv2D(feat_2, 96, 3x3, stride = 2, relu) 17. feat_3 ← AttentionFusion(p2, x3)
Step 5: Secondary feature path from input	18. y ← Conv2D(input, 64, 1x1, stride = 2, relu) 19. y ← Conv2D(y, 64, 3x1, relu) 20. y ← Conv2D(y, 64, 1x3, relu) 21. y ← MaxPool(y, 3x3, stride = 2) 22. y ← Conv2D(y, 64, 1x1, relu) 23. y ← Conv2D(y, 64, 5x1, relu) 24. y ← Conv2D(y, 96, 1x5, relu) 25. y ← Conv2D(y, 96, 1x1, relu) 26. y ← Conv2D(y, 96, 3x3, relu) 27. y ← MaxPool(y, 3x3, stride = 2) 28. feat_final ← AttentionFusion(feat_3, y)
Step 6: Classification head	29. flat ← Flatten(feat_final) 30. fc1 ← Dense(flat, 1024, relu) 31. fc1 ← Dropout(fc1, 0.4) 32. fc2 ← Dense(fc1, 256, relu) 33. fc2 ← Dropout(fc2, 0.4) 34. output ← Dense(fc2, 2, softmax)
Return	Model (input, output)

MPFAN-AF, multiscale parallel feature aggregation network with attention fusion.

The MPFAN-AF model is implemented using Python with TensorFlow and Keras DL libraries to facilitate reproducibility and potential clinical translation. All input images are resized to 120×120 pixels and normalized to the [0, 1] intensity range. Data augmentation techniques such as horizontal and vertical flipping, rotation, and brightness adjustment are applied during training to improve generalization and robustness. The model architecture comprises parallel convolutional pathways with varying kernel sizes (e.g., 1×5, 5×1, 3×3) to extract multi-scale spatial features, which are subsequently refined and fused using attention mechanisms, as detailed in Table 1 and 2. The model is trained using the Adam optimizer with a learning rate of 0.001, a batch size of 16, and categorical cross-entropy loss. Dropout regularization with a rate of 0.4 is incorporated after dense layers to mitigate overfitting. The

complete cataract detection pipeline comprises image preprocessing, feature extraction via the MPFAN-AF network, and final classification into normal or cataract classes through a softmax-activated output layer. In order to facilitate transparency and the uptake, the source code of the full implementation, including pretrained model weights and specific usage guidelines, will be publicly released in a GitHub repository following publication acceptance.

Results and discussion

The experiments have been conducted on the device, which consists of an Intel i7 9th generation Processor, a NVIDIA GTX 1650 GPU, 8GB RAM and 256GB SSD storage. This mid-range specification emphasizes the fact that the model can be used on readily available hardware, thus adding to its relevance in real-life and limited-resource scenarios. The proposed MPFAN-AF model is assessed against a publicly available dataset dedicated to the tasks of cataract classification.

Cataract eye dataset

The Cataract Image Dataset for this study is obtained from Kaggle and authored by Nandan Padia [23]. This dataset aims to aid in the development of cataract detection DL algorithms that analyze eye images directly, rather than relying on textual medical records or clinical files. It poses itself as a great asset for computer vision endeavours concerned with recognizing cataracts from photographs. It has 612 labelled eye images distributed into training and testing subsets to facilitate supervised learning and model evaluation. Each image is labelled as either Cataract or Normal (non-cataract); therefore, the dataset can be used for image classification tasks.

As illustrated in Table 3, the dataset is balanced across both classes with an equal distribution of cataracts and normal images (306 each). This balance supports a constructive competition among various models trained on the same data, which helps improve a self-contained comparison of the model's efficiency. The images are stored as .jpg files and are eligible for preprocessing such as resizing, normalization, and data augmentation, which are standard within DL frameworks. The current dataset is a good initial resource in terms of training CNN and other vision-based networks that will be used in cataract detection and classification in automated mobile healthcare environments.

Figure 4 shows a visual contrast between the eyes affected by cataract and a normal eye. The existence of visible opacities and clouding or discolouration in the lens region presents the critical classification characteristics to the models. Natural eyes do not contain blacks as they possess clear eyes with an even appearance across the entire lens set-up, which explains why feature discrimination is important in the proposed DL system.

Experimental outcomes and analysis

This section of the experimental findings examines the capability of the recommended MPFAN- AF model when it comes to diagnosing cataracts. Evaluation measures covered the evaluation measure, accuracy, precision, recall, F1-score loss measures in training and testing data sets [17, 19, 24]. Results indicate that this model offers robustness, effective classification results, and outstanding generalization capability for unknown data. A complete performance trend analysis accompanies the discussion of how well the model works and maintains stability throughout training.

The accuracy data shown in Figure 5a–5e presents a representation in Figure 5a that demonstrates the model achieving stable accuracy levels above 95% while exhibiting quick early growth during the 100 epochs. This steady improvement indicates effective learning of discriminative features from fundus images. The proximity of training and test accuracy curves suggests minimal overfitting and strong generalization. Figure 5b, which shows the loss curves, reveals that while training loss continuously declines, the test loss displays higher variance and a sudden spike around epoch 55. This fluctuation may be attributed to over-sensitive weights or momentary instability during optimization. Despite this, the loss curve re-stabilizes, and the overall decreasing trend confirms the convergence and resilience of the model to temporary irregularities.

Figures 5c–5e provide deeper insight into the classification reliability of the model. Figure 5c presents the precision curves, where both training and test values stabilize above 95%, indicating that the model maintains a low false-positive rate across training and validation data. Figure 5d shows recall trends, again with values exceeding 90% for both training and test sets, highlighting the model's capability to detect actual cataract cases with few false negatives, which is particularly crucial in medical diagnostics. Figure 5e combines the outcomes of precision and recall in the form of F1-score, where both curves remain consistently high (close to 1.0), underscoring the model's balanced performance. These combined results illustrate that MPFAN-AF learns effectively and maintains its performance when applied to unseen data, confirming its potential as a robust and reliable tool for automated cataract classification.

Table 3 Cataract eye dataset distribution

Subset	Cataract Images	Normal Images	Total Images
Training	245	246	491
Testing	61	60	121
Total	306	306	612



Figure 4 Sample of images for each class of the cataract dataset.
Example images from the Cataract Image Dataset (Nandanp6. Cataract Image Dataset. Kaggle. Accessed March 15, 2025. Available at: https://www.kaggle.com/datasets/nandanp6/cataract-image-dataset).

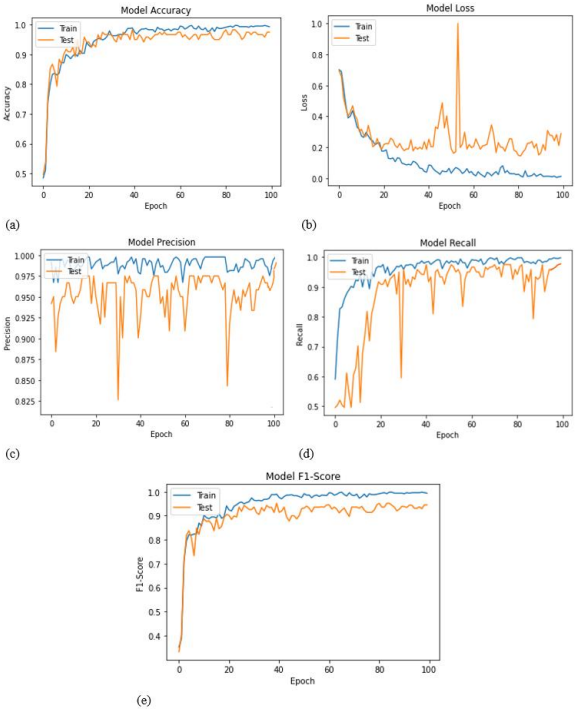


Figure 5 The trade-off between training and validation groups on (a) accuracy, (b) loss, (c) precision, (d) recall and (e) f1-score curves.

The confusion matrix in Figure 6 illustrates the classification performance of the MPFAN-AF model on the cataract detection task, revealing a high degree of accuracy with 97.52%. Out of 61 actual normal samples, 59 were correctly classified with only two misclassifications, while 59 out of 60 cataract images were accurately identified, with just one false negative. This strong diagonal dominance confirms the model's robust ability to distinguish between cataract and non-cataract classes. The minimal occurrence of false positives and negatives reflects excellent sensitivity and specificity, critical in medical diagnostics to avoid misdiagnosis. The results align with the model's previously demonstrated high precision (up to 98.33%) and recall (up to 96.72%), underscoring its effectiveness and generalizability for real-world clinical deployment.

The radar charts, as shown in Figure 7, illustrate the model's classification performance visually and support a deeper analytic comparison between Class 0 and Class 1 using key metrics. From an analytical standpoint, both classes show excellent results in terms of true positive rate (TPR) and true negative rate (TNR), above 96, indicating the model's strong ability to identify both positive and negative samples correctly. Class 1 edges slightly ahead in TPR (0.9831 vs. 0.9672), meaning it is more sensitive in detecting true positives. On the other hand, Class 0 slightly outperforms Class 1 in True Negative Rate (0.9831 vs. 0.9672), showing a better rate of correctly rejecting negatives. This trade-off between sensitivity and specificity highlights the model's internal balancing of classification thresholds, which may be optimized depending on whether the application values minimizing false negatives or false positives more. The false positive rate (FPR) and false negative rate (FNR) values are very low across classes, but Class 1 has a higher FPR (0.0328) and lower FNR (0.0169), while Class 0 is the reverse. Analytically, this indicates Class 1 is more aggressive in labeling positives—catching more true cases but at a slight cost of increased false alarms.

Further analysis of the false discovery proportion (FDP) values (Class 0: 0.0169, Class 1: 0.0328) reveals that while Class 1 predicts more positives, it also has a slightly higher proportion of them being incorrect. This metric aligns with the slightly elevated FPR in Class 1 and suggests reviewing decision thresholds if false positives are costly in the application context. The identical total test number (TTN = 59) for both classes ensures fairness and statistical consistency in this comparison, removing the influence of sample imbalance. From a holistic analytical perspective, the radar plots confirm the model's reliability, with metrics tightly grouped near ideal values. However, the minor performance differences between classes could be further analyzed using precision-recall trade-offs to fine-tune the classifier, especially for real-world scenarios where the cost of errors varies by class. These findings indicate that the model is well-optimized but leaves room for application-specific adjustments.

Figure 8 displays the prediction output of the MPFAN-AF model on the various test eye image instances. The Figure showcases how well the model deals with problems, including variations in eye colour, lighting, makeup, and optical obstructions. These problems pose significant challenges in the classification of diseases in practice using images. The upper row and lower row illustrate the properly classified examples of the model, which depicts the comprehensiveness of the features in texture, contrast and iris features captured by the model to reflect that it effectively extracted features and self-attention to essential characteristics. In the model, the invariance is also impressive in other visually stimulating cases, such as chaotic reflections, heavy makeups, and other mixed patterns, portraying it to have a robust and generalization ability even under changing environments. The comparison demonstrates the reliability of this model in practical situations where there are extremely noisy images, such as medical screening, or some simplified cases, like biometric recognition, where there is a lot of variation within a particular group or a particular class.

The model of MPFAN-AF suggested is highly effective in real-time, especially in cases with images taken by mobile cameras or downloaded from the internet. As indicated in Figure 9, the model is able to render accurate predictions even on non-standardized inputs that constituted adaptations in lighting, focus, and background noise provided in the non-standardized input, similar to real-world deployments of the model. This reveals the resilience of the model and a high generalizing performance outside the regulated or clinical settings. The ability to maintain high accuracy on such diverse and low-resource inputs makes MPFAN-AF especially suitable for mobile-based screening tools, telemedicine platforms, and community-level eye health monitoring, where imaging quality may vary significantly. These results validate the model's potential for scalable, field-ready cataract detection in both urban and rural healthcare settings.

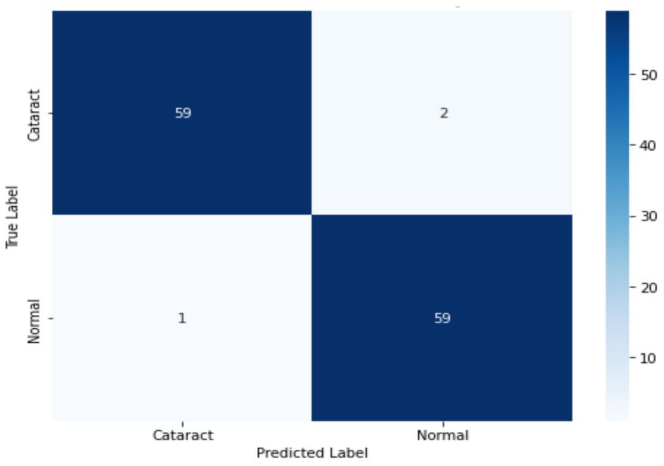


Figure 6 Confusion matrix of proposed MPFAN-AF for cataract classification.
MPFAN-AF, multiscale parallel feature aggregation network with attention fusion.

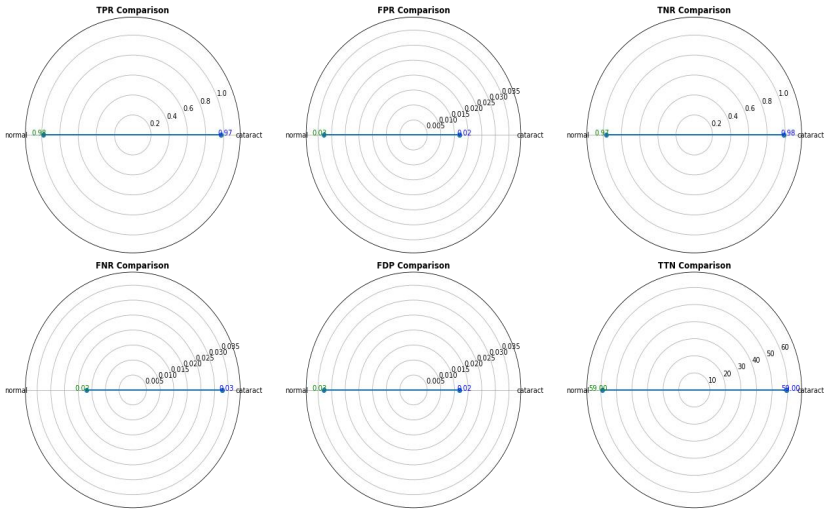


Figure 7 Spider (radar) curve showing other measurements for MPFAN-AF.
MPFAN-AF, multiscale parallel feature aggregation network with attention fusion; TPR, true positive rate; TNR, true negative rate; FPR, false positive rate; FNR, false negative rate; FDP, false discovery proportion; TTN, total test number.



Figure 8 MPFAN-AF predicting model behavior over test instances.
MPFAN-AF, multiscale parallel feature aggregation network with attention fusion. Example images from the Cataract Image Dataset (Nandanp6. Cataract Image Dataset. Kaggle. Available at: https://www.kaggle.com/datasets/nandanp6/cataract-image-dataset. Accessed 15 Mar 2025.)

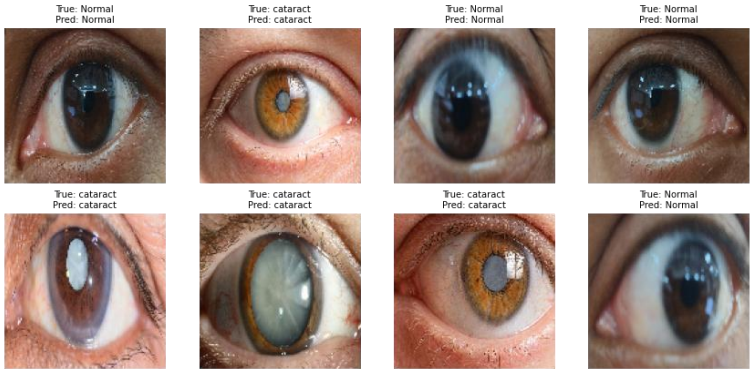


Figure 9 MPFAN-AF predicting model behavior over real-time instances.
MPFAN-AF, multiscale parallel feature aggregation network with attention fusion. Representative clinical photograph of a patient. Written informed consent for publication of this image was obtained using the World Health Organization (WHO) "Informed Consent Form for Clinical Research".

Comparison with existing methods

Three-line graphs in Figure 10 demonstrate that the MPFAN-AF model (cyan) outperforms VGG16, InceptionV3, ResNet-50, DenseNet-121, and MobileNet for all performance metrics (accuracy and F1-score) during 100 epochs of training, thus establishing its effectiveness for cataract classification. All baseline models are initialized with pretrained weights from ImageNet and subsequently fine-tuned on the cataract dataset to ensure a fair and optimized comparison. This evaluation strategy underscores the superior classification capability and generalization performance of the MPFAN-AF model in cataract detection tasks.

The analysis of models reveals that each of them differs from one another in precision, recall, and accuracy. As far as the case of conventional DL models is concerned, VGG16, Inceptionv3, Xception, and ResNet-50 offer reasonable performance, though they still lag newer architectures. For example, ResNet-50 and Xception boast an accuracy of 93.39%, while VGG16 and InceptionV3 trail a bit at 90.08% and 91.73%, respectively. Although these models are useful, their lower precision and recall scores suggest that they achieve some level of balance between true positive detection and false positive minimization. On the other hand, DenseNet-121 and MPFAN pose a substantial increase in recall, achieving 96.00% and 96.80%, respectively. Yet, MPAN-AF (Proposed) surpasses all other models with exceptional performance in all metrics by showing 98.33% precision, 96.72% recall, and 97.52% accuracy. These numbers display that the proposed model can reduce the number of false positives the model gives out compared to the number of true positives, making precision, recall, and accuracy all remarkably high, proving useful in practical scenarios that demand trustworthiness.

In the analysis, the performance of the MPAN-AF (proposed) model is attributed to its feature architecture and extraction, which are well executed. It outperforms other models, such as VGG16, ResNet-50 and Xception, by 7.63% (90.70% to 98.33%) in accuracy and by 4.13% (93.39% to 97.52%) in precision. Specifically, the superior performance is attributed to its multiscale convolutional structure, which captures features at various levels of granularity, and the attention fusion mechanism, which allows the model to focus on the most relevant regions of the image. These architectural advantages enable MPFAN-AF to achieve higher accuracy, precision, and recall compared to standard CNN models. Further ablation studies confirm that important design changes, such as adding attention features and changing the optimized feature layers, aid the performance. For example, attention layers are shown to reduce both precision and recall by a large margin, thus proving their importance for enhancing the model performance. These results indicate that MPAN-AF (proposed) achieves a balanced level of accuracy, recall and precision while outperforming other models, which is crucial for applications focusing on human activity detection and security systems due to the required reliability and efficiency.

The dual-axis plot in Figure 11 provides a comparative view of trainable parameter counts (log-scaled) and classification accuracy across a range of benchmark DL models used for cataract classification. Traditional architectures such as VGG16, InceptionV3, Xception, and ResNet-50 possess large parameter sizes ranging from 22.9M to 138M, yet yield only moderate accuracies between 90.08% and 93.39%. While these models are known for their strong feature extraction capabilities, their computational burden makes them less suitable for lightweight or real-time deployment scenarios. DenseNet-121, with a reduced footprint of 8M parameters, achieves 94.21% accuracy, showing improved efficiency through dense connections. EfficientNet-Lite0 (5.1M) and MobileNetV3-Large (3.35M) offer even smaller architectures with accuracies of 94.80% and 94.50%, respectively, indicating their suitability for edge computing.

Although MobileNetV3-Large has fewer parameters (3.35M) compared to MPFAN-AF's 4.64M, it is important to note that MPFAN-AF significantly outperforms MobileNetV3-Large and all other models in terms of accuracy, achieving a remarkable 97.52%, the highest among all evaluated architectures. This demonstrates that MPFAN-AF maintains a compact model size and incorporates architectural innovations, such as multiscale parallel feature aggregation and attention fusion. It yields superior feature representation and classification capability. Furthermore, while its predecessor MPFAN achieves 95.37% accuracy with 42.5M parameters, MPFAN-AF delivers better performance with nearly 9× fewer parameters, reinforcing its suitability for mobile, embedded, and telemedicine applications where both efficiency and accuracy are critical. The trend in Figure 11 clearly validates MPFAN-AF's outstanding balance between computational cost and diagnostic precision.

Ablation study

We tested the significance of MPFAN-AF architecture components through systematic testing, which removed essential modules or partially altered them. Table 4 shows the ablation study of this proposed model. The research protocol employed an identical experimental configuration to the complete model, with a learning rate set to 0.001 and a batch size of 16, along with 100 training epochs. A series of evaluated components includes:

1. Baseline CNN. Standard CNN without multiscale convolutions, parallel aggregation, or attention fusion.
2. The MPFAN without Attention Fusion variant maintains all multiscale parallel aggregation components yet omits attention fusion modules.
3. The MPFAN variant uses only 3×3 convolution kernels to eliminate the multiscale characteristics from the network.
4. MPFAN-AF (Proposed). Full model with multiscale convolution, parallel aggregation, and attention fusion.

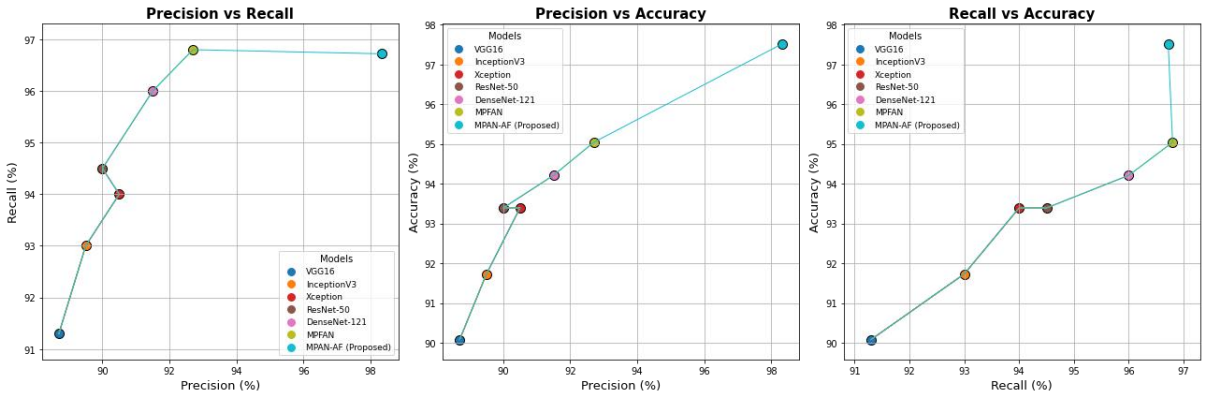


Figure 10 Comparative performance of MPFAN-AF and other models for cataract classification.
MPFAN-AF, multiscale parallel feature aggregation network with attention fusion.

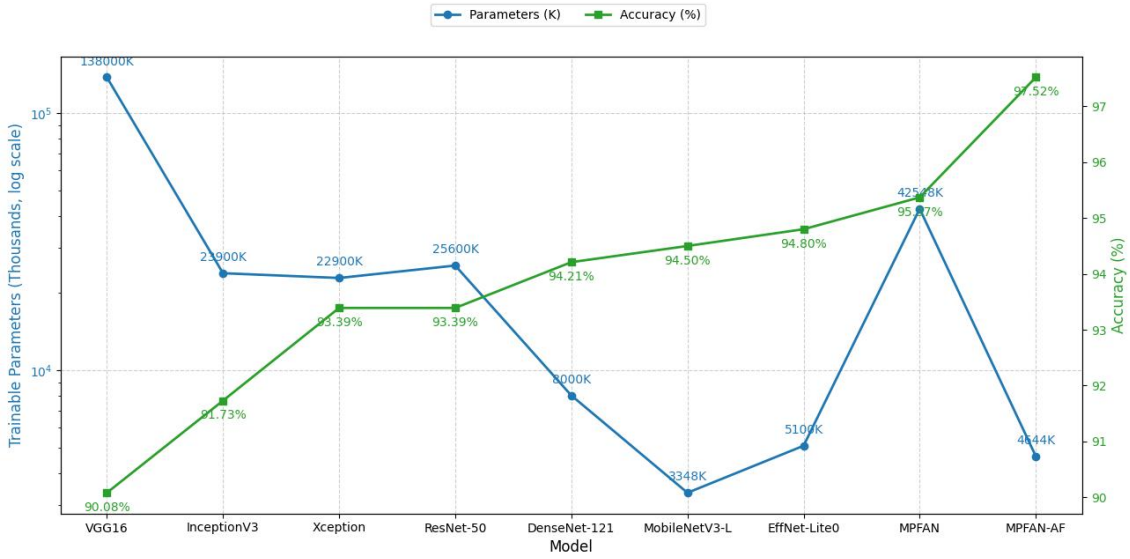


Figure 11 Comparison of model size and accuracy.

Table 4 Performance comparison of model variants on tomato disease classification

Model variant	Accuracy (%)	Precision (%)	Recall (%)
Baseline CNN	90.74 ± 0.42	91.10 ± 0.38	90.16 ± 0.51
MPFAN w/o Attention Fusion	94.21 ± 0.36	94.59 ± 0.40	93.44 ± 0.47
MPFAN with Single-Scale Convolutions	95.37 ± 0.33	95.84 ± 0.29	94.26 ± 0.35
MPFAN-AF (Full Model)	97.52 ± 0.27	98.33 ± 0.22	96.72 ± 0.31

CNN, convolutional neural network.

The attention mechanism and fusion modules considerably eliminate multitasking with branching attention fusion over the network. It increases value capture across the timing and frequency domains in the proposed system over single-channel attention fusion using dual-scale FFT attention. In contrast to before, the attention fusion boosted accuracy from 94.21% back to 97.52%, with precision increasing to 98.33% and recall escalating to 96.72%. With single-scale convolutions, maintaining attention fusion, partial boosts allowed precision to reach 95.84% and recall 94.26%. MFPAN demonstrates its performance and enhances the model's ability to extract patterns of cataract at different layers of the multiscale level using multiscale feature extraction. The baseline CNN utilizes a sequential design without incorporating attention, which shows the worst performance out of all at 90.74% accuracy, 91.10% precision, and 90.16% recall, highlighting the underperformance of stagnant architectures in imaging. Using multiscale spatial frequency filtering defined by 1×1, 3×3, 5×1 and 1×5 filters, the model can learn the local fine textures and coarse level distortions at the spine of cataracts. These alterations offer MFPAN alongside better visual clarity and improved comparability and compatibility for users and agents using the parallel feature aggregation, keeping the pulled features unique without merging the spatial information. Healthcare-relevant regions, such as lens opacities, along with structural blurring, receive emphasis through the attention fusion mechanism based on global average pooling with dense layers implementing sigmoid activation. The attention mechanism is essential in detecting cataract cases because its elimination leads to a significant recall decrease from 96.72% to 93.44%. The precise diagnosis of cataracts depends on combining multiscale aggregation with attention-based refinement to reach better technical performance and minimize false negative outcomes in clinical practice.

Each model variant is trained and evaluated over five independent runs (5-fold) using identical hyperparameter settings to ensure statistical reliability. Table 4 reports the mean and standard deviation

values for accuracy, precision, and recall. Standard deviation is computed using the standard population formula to measure the variability of each metric across the five runs. These metrics allow for assessing the stability and conformance of each of their performance. All variants of the MPFAN-AF model demonstrate low standard deviations compared to other variants, indicating that classification has a high and stable performance on varied training instances.

Conclusions and future directions

This paper proposes the MPFAN-AF model for cataract classification using eye images, integrating multiscale parallel feature extraction with an attention fusion mechanism. This design enables the model to focus on both diverse spatial features and critical diagnostic regions, effectively addressing the limitations of conventional CNN-based approaches. Experimental results demonstrate that MPFAN-AF significantly outperforms existing architectures, achieving substantial precision, recall, and overall accuracy gains. Our ablation studies further validate the impact of each architectural component, particularly the attention fusion module, which enhances the model's ability to focus on clinically relevant features.

Despite its advanced architecture, MPFAN-AF maintains a lightweight design with an optimized parameter count and low memory footprint, enabling fast inference on standard hardware. This efficiency makes it suitable for integration into mobile or edge devices, supporting real-time cataract screening in telemedicine and resource-constrained clinical settings. The model is compatible with standard imaging formats and can be easily embedded into hospital information systems or portable diagnostic platforms as a clinical decision-support tool. By delivering automated and accurate preliminary assessments, MPFAN-AF has the potential to streamline patient triage, reduce diagnostic delays, and improve access to eye care.

However, the current model is trained on a single-source dataset, which may introduce potential biases due to limited demographics, imaging conditions, and equipment variability. This could affect its generalizability to broader clinical populations. To address this, future work will focus on collaborating with hospitals and medical institutions to access larger, more diverse datasets. Additionally, techniques such as domain adaptation and fairness-aware learning will be explored to reduce bias and ensure equitable performance across different patient groups.

We also recognize the challenges of deploying the model in real-world settings, where images may suffer from poor lighting, motion blur, occlusion, or low resolution. Future research will incorporate advanced data augmentation, image enhancement preprocessing, and training on real-world clinical data to improve robustness under such conditions. Moreover, extending MPFAN-AF for multi-class classification of various ocular diseases—such as different cataract types, glaucoma, diabetic retinopathy, and macular degeneration—is a promising direction that can broaden its diagnostic utility.

Given its compact architecture and strong generalization capabilities, MPFAN-AF is well-positioned for deployment on portable devices and mobile platforms. Future development may focus on optimizing the model for low-power environments to enable real-time, point-of-care diagnostics. Integration into remote screening software and validation through clinical collaborations will help ensure real-world applicability and compliance with regulatory standards. Additionally, incorporating explainable artificial intelligence techniques will further enhance the interpretability and clinical trustworthiness of the model.

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